TRACE-RPG: Auditable Proposal-to-Commit Boundaries and Source-Bound Read-Only Ontology Observation in Interactive Games

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***Abstract*—Generated game events can satisfy a schema and cite known facts while requesting inaccessible objects, unauthorized state changes, or premature disclosure. TRACE-RPG separates proposal authority from commit authority: an external deterministic gate checks the encoded state and authored policy, supplies typed errors for repair, and revalidates the final candidate. Transition records support replay and assignment-complete accounting. A source-bound bilingual ontology view exposes these decisions without granting write access.**

The deterministic audit crosses 32 graph/state combinations with 13 candidate variants, assigning 416 cases to each of four arms. Against an implementation-separated reference, the schema-only control committed 260/416 reference-invalid final candidates; each semantic-gate arm had 0/416 reference-invalid final commits. Guided repair recovered 128/260 initially invalid proposals into reference-valid commits and retained all 11/11 initially valid structural signatures. Reachability closure and breadth-first search agreed on 32/32 graph/states. In paired sentinels, six authored undeclared paraphrases passed the structured semantic gates, while six counterparts explicitly declaring forbidden facts were held. This contrast identifies a declared-field coverage limit, not a natural-language safety rate. The evidence covers authored finite reusable-key graphs and implementations that share encoded rules, not independently validated ground truth or language-model or human efficacy.

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

GENERATED events must satisfy a game's data contract and current rules. A proposal may require an item inside a locked room, disclose a fact the NPC knows but may not yet reveal, or perform a permitted action that does not advance the objective. Grounding and plausible dialogue [1], [2] supply content; the game must still decide which state changes to allow.

TRACE-RPG enforces this decision at a proposal-to-commit boundary (Fig. 1). Models and adapters submit typed candidates. The external deterministic gate checks preconditions, object reachability, NPC knowledge, disclosure permission, effects, and quest-stage constraints. Failed checks return structured repair information. Repair changes the candidate, not the state; revalidation against the unchanged state and policy authorizes a commit. The read-only ontology view links source declarations to checks and public decisions. Replay and assignment records preserve the resulting execution path. The research question is which checks distinguish a well-formed proposal, an authorized transition, and an achieved objective, and which records support each decision.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1	frozen offline repairability classes

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

In v5, ten of 20 assigned responses passed parsing, eight passed the semantic gate, and seven reached the goal: authorization did not imply objective success. Ordinary-rule and ontology-gate implementations agreed on 60/60 cases under frozen disclosure rules. The deterministic audit in Sec. V-B tests a schema-only intervention and a separated reference, recording proposal–repair–hold outcomes across a finite graph family.

Table I links the integration's contracts to their evidence: **C1—contracts**, **C2—validation/repair**, **C3—zero-regression audit binding**, **C4—assignment accounting**, and **C5—guided repair**. Each evidence lane retains its own denominator and inference boundary.

II. Related Work and Research Gap

Grounding, memory, planning, and role-based agents [1], [2], [3], [4] and player/dialogue systems [5], [6], [7], [8] support contextual interaction. We examine whether the current encoded state and disclosure policy permit a proposed transition. Affect and engagement [9], [10], [11] require different outcomes. Style preferences, affect estimates, and personality labels may guide

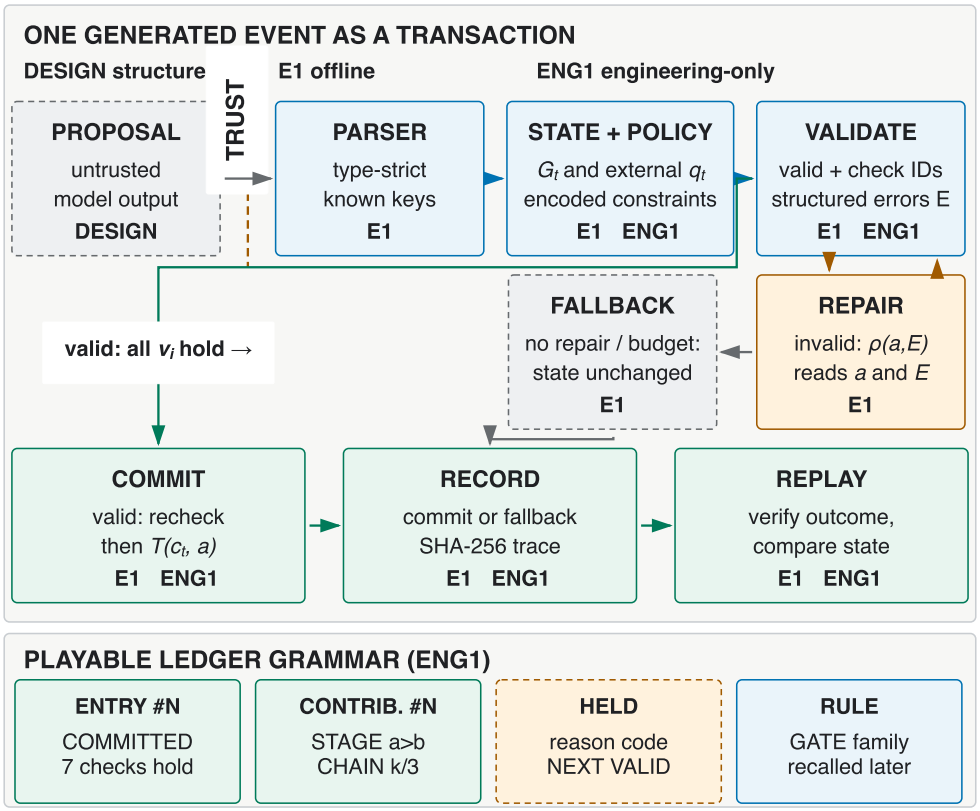


Fig. 1. Policy/c_t gates proposals; errors feed bounded ρ or unchanged fallback. Passing candidates undergo revalidation, commit, recording, and replay. Historical cohorts and legacy ENG1 remain archived, not active model evidence. The lower panel shows the committed-snapshot ledger grammar.

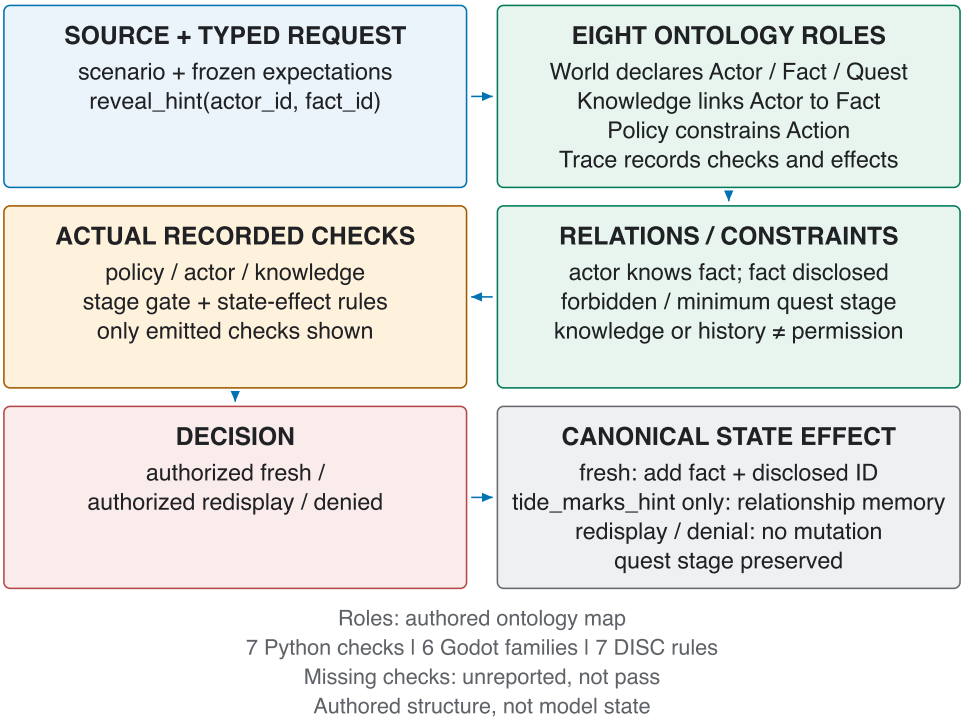


Fig. 2. The eight-role ontology structure (Table III) links source fields to checks and observed decisions. Authorized transitions alone change canonical state. This source diagram is neither a live capture nor hidden-state evidence.

generation but do not authorize actions. MBTI questionnaire alignment alone would not establish NPC voice stability or a causal personality benefit; neither outcome is measured here.

Existing benchmarks and automated play-testing cover state, embodiment, content, process, social interaction, and long-horizon play [12], [13], [14], [15], [16], [17], [18], [19], [20]. NCP-Bench measures narrative commitment/fact consistency, not player outcomes [21]. Structural/schema, affordance, and rule methods motivate our constraints [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33]; their use does not establish superiority. Retrieval and memory provide context, not commit authority [34], [35], [36].

Our reporting follows work on frozen artifacts, retained failures, construct/rater/bias controls, and justified margins/random effects [37], [38], [39], [40], [41], [42], [43], [44]. Energy and carbon reporting motivates resource accounting [45]. Replay and leakage checks support audit records and denominators, not efficacy [46], [47]. C1–C5 integrate established enforcement, transaction management, feedback repair, action interposition, preconditions/effects, abort, social/belief-aware planning, and replay mechanisms [27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]. We claim an auditable integration with explicit evidence boundaries, not invention or first combination of these algorithms.

III. Problem Definition and Guarantee Boundary

In $c_t = (G_t, q_t)$, G_t is the typed game snapshot and q_t contains externally authored action-policy, disclosure, and quest-stage constraints. Python carries both in `WorldState`: locations, reachable locations, object locations, inventory, facts, NPC knowledge, and quest stage describe the state; action policies and forbidden disclosures constrain it. An `ActionPolicy` specifies required preconditions, allowed and required effects, and allowed quest-stage targets. These are two conceptual roles, not independently stored Python objects. Predicates neither read $h_{<t}$ nor reconstruct `WorldState`. Transitions update G_t only and exclude policy rotation. Retrieval, scores, rationales, memory, and future affect have no mutation authority.

After strict parsing of an untrusted proposal a_t, the validator runs seven deterministic predicates and returns `ValidationResult(valid, errors, checks_run)`:

$$\begin{aligned} V(c_t, a_t) &= (\text{valid}, E_t, C_{\text{run}}), \\ \boldsymbol{v} &= (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, \\ &\qquad v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \end{aligned} \tag{1}$$

Here E_t contains structured errors with code, entity, reason, and repair hint; C_{run} lists the seven executed check identifiers. The aggregate $\text{valid} \equiv \bigwedge_{i=1}^{7}(v_i = 1)$ summarizes the seven predicate outcomes: policy, precondition, reachability, NPC knowledge, forbidden disclosure, stage eligibility, and stage monotonicity. The runtime returns `valid` and check identifiers, not seven individual Booleans; an absent per-check trace or omitted field remains unobserved. The transition below applies to the current candidate a_t, which repair may have changed, rather than the original proposal's entire transaction. Its $\boldsymbol{v}$ checks use unchanged c_t. Rejection leaves c_t unchanged while bounded repair or fallback continues; passing same-policy revalidation authorizes mutation:

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

Execution follows proposal → deterministic validation → observable validity, check identifiers and structured errors → optional bounded repair → same-state revalidation → commit or unchanged fallback. A candidate is checked again immediately before application; repair exhaustion or an absent repairer leaves state unchanged. The guided operator uses only the proposal and errors. The general Python callback also receives `WorldState`, so the restricted-input claim does not cover every callback. A controlled comparison must give deterministic and model-based detailed repair the same proposal and errors; the same model's generic-error condition receives a generic signal. This design alone implies no comparison outcome.

The fixtures test invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \in \mathbb{Z}_{\geq 0}$ bound attempts to $K + 1$ with pre-commit revalidation; without deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** a recorded initial snapshot, complete ordered transition record, and frozen schema and deterministic transition version permit reconstruction of the recorded terminal structured state. Reconstruction does not authenticate the record, establish semantic correctness, or reproduce model generation.

These invariants assume correct policy/extraction, deterministic versions, returning callbacks, and single-process ownership. Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

A. Schema Validity, Authorization, and Goal Success

Fix the encoded pre-action state c_t, authored policy, and strict candidate schema. Let $\mathcal{S}$ contain schema-valid candidates; define authorization and objective achievement separately:

$$\begin{aligned} \mathcal{A}(c_t) &= \{a \in \mathcal{S} : V(c_t, a).\text{valid}\}, \\ \mathcal{G}(c_t) &= \{a \in \mathcal{S} : \text{Goal}(T(c_t, a))\}, \\ \mathcal{H}(c_t) &= \mathcal{A}(c_t) \cap \mathcal{G}(c_t), \\ \mathcal{H}(c_t) &\subseteq \mathcal{A}(c_t) \subseteq \mathcal{S}. \end{aligned} \tag{3}$$

The set $\mathcal{G}$ includes candidates with a defined encoded transition T; evaluating that transition does not authorize execution. By definition, $\mathcal{H}$ requires both authorization and goal attainment, and authorization requires schema validity. A schema-valid forbidden-fact declaration can fall outside $\mathcal{A}$. A permitted `observe` action can fall inside $\mathcal{A}$ but outside $\mathcal{H}$ if it leaves the goal false. Neither $\mathcal{A} \subseteq \mathcal{G}$ nor $\mathcal{G} \subseteq \mathcal{A}$ follows in general. This conditional set argument is not a new verification theorem: it assumes faithful state/policy extraction, correct predicates and

TABLE II
ENCODED VALIDATOR BOUNDARY

Predicate	Rejects	On failure
Action policy	unknown action/type; omitted required preconditions/effects; disallowed effects or unauthorized quest-stage effects	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	used/disclosed fact absent from actor knowledge	unchanged
Disclosure	forbidden fact	unchanged
Quest stage (`quest_stage`)	required stage above current	unchanged
Quest-stage monotonicity (`quest_stage_monotonicity`)	regressive stage	unchanged

TABLE III
EIGHT ONTOLOGY ROLES WITH READ-ONLY SOURCE BINDINGS; THE MAP PERFORMS NO LIVE INFERENCE AND GRANTS NO AUTHORITY

Role	Source declaration or relationship example
World	Initial object locations place the signal lens in the lamp store.
Actor	Captain Mira is the sole eligible actor.
Knowledge	Actor knowledge refers to canonical Fact IDs, not permission.
Fact	Initial facts include `dock_saved` and `lighthouse_dark`; `keeper_betrayal` is an ID, not payload.
Policy	Disclosure policy forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2.
Quest	The initial quest is at stage 0 with flag `dock_saved`; disclosure preserves stage.
Action	`reveal_hint(actor_id,fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged.
Trace	Prior and next state hashes, rule and template identifiers, and case ID bind packet/version to evidence; absent fields cannot establish PASS.

transitions, and an unchanged pre-action snapshot. Strict parsing alone cannot establish these state-dependent conditions. We report no XGrammar execution or token-masking comparison.

Knowledge and disclosure permission are separate inputs. In the audit, the NPC knows `keeper_secret` at both stages, but disclosure is forbidden at stage 0 and permitted at stage 1. Permission changes without adding knowledge. Retrieval, rationale, affect, style, and personality scores cannot override this policy.

IV. TRACE-RPG ARCHITECTURE AND ALGORITHMS

A. Trust Boundary and Contracts

Models and adapters propose actions; the parser rejects ambiguous types and unknown keys; externally authored policy defines preconditions and effects. The versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1). Unknown-key rejection parity proves neither parser equivalence nor portability. Python runs exactly seven deterministic checks: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. Godot uses six state-relative families and seven DISC rules; these labels are not isomorphic. Its `QUEST STAGE` display spans two Python quest checks.

The ontology role map links source declarations to frozen expectations. Table III's `reveal_hint(actor_id,fact_id)` distinguishes observation, repair, and authority. The view displays the `tide_marks_hint`/stage-2 policy. The repair operator ρ receives only a_t, E_t, may insert a named predicate, and cannot read `WorldState`. Godot commits only all-seven-pass candidates. Fresh disclosure increments revision $+1$; redisplay does not.

The gate checks declared `used_facts` and `disclosed_facts`; it does not infer fact IDs from `narrative_text`. Prose does not directly mutate canonical facts or quest stage. NPC-DISC-1 selects approved disclosure/denial templates, a narrower contract than unrestricted generated dialogue. An undeclared paraphrase can still disclose a secret to a reader. The paired sentinels in Sec. V-B test this declared-field boundary, not an NLI detector or renderer-level safety mechanism.

1) Zero-regression audit binding (C3): The eight-role contract links declarations, executed checks, decisions, state effects, and replay receipts without granting the observer mutation authority. Under frozen NPC-DISC-1 rules, `general_rule` and `ontology_gate` agree on 60/60 outcomes, with zero observed expectation, state, display, or replay mismatches in either arm (Tables XIII–XV). This declaration-derived agreement is a bounded zero-regression audit binding, not held-out safety or ontology superiority. TR01–TR06 remains `PROPOSED_NOT_VALIDATED`; no traceability score or comparative gain was measured. NPC-DISC-1's authored multi-state fixture covers policy, actor, stage, boundary, and malformed-input cases. Its 60 cases form a rule-conformance matrix, not 60 independently sampled game episodes.

B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims and plans, and Godot alone writes state. The harbor-restoration/authorized-tide-lead goal is fixed, not model-editable. Bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. The service uses `127.0.0.1:8768`; allowed browser origins are `localhost` and `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1 contains 60 declaration-derived `reveal_hint` cases; it does not rerun or enlarge historical `verified-run-006`. Results and Tables XIII–XV specify strata, bindings, and input-to-check mappings. Godot rechecks service bindings before mutation.

C. Validate–Repair–Commit

Fig. 3 gives dependency-free pseudocode with repair budget $K \in \mathbb{Z}_{\geq 0}$. Record appends in memory; persistence

$a \leftarrow \text{Propose}(c)$; $j \leftarrow 0$
repeat
  $(\text{ok}, E) \leftarrow V(c, a)$; $\text{Record}(a, E)$
  if ok **then**
    assert $V(c, a).\text{ok}$; **return** $\text{Commit}(c, a)$
  if $j = K$ **then return** $\text{Fallback}(c)$
  $a \leftarrow \text{Repair}(c, a, E)$; $j \leftarrow j + 1$
until terminal

Fig. 3. Validate–repair–commit records at most $K + 1$ candidate attempts and revalidates a passing candidate before mutation.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	Edit $\delta(e)$ on the candidate
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate edit would launder or needs state)

requires terminal consistency. (ok, E) abbreviates V, with $\text{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c, a).\text{ok}$ performs revalidation.

Repair cannot mutate state, and exhaustion preserves it. The typed `counterexample_guided_repair` operator receives candidate a and errors E, its sole state-derived input, never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{4}$$

In Table IV, $\oplus$ applies deterministic edits $\delta(e)$ to one field/error entity, with at most $|E|$ edits per attempt. Conflicting requirements and rejections leave fields unchanged. A fixed point requires identical E; it does not establish convergence. The nondeployable state-reading oracle ignores errors and reconstructs preconditions, effects, and both quest-stage fields from state. It never removes required objects, used facts, or disclosed facts to conceal a violation.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 detects accidental and tested changes, not identity or coordinated rewriting. Replay checks the prior state, candidate, validation, invalid nonfinal attempts, recomputed final state, and JSONL continuity; it does not replay repair generation. Request→decision→result and first-action/one-receipt bindings do not prevent duplicate IDs or impersonation. Terminal classes are commit, symbolic fallback, adapter failure, and controller failure; failures preserve state/assignment without a completed proposal trace. Pre-execution run/arm/scenario/seed/model/revision keys and configuration/input/prior-state hashes bind assignments. Aggregation rejects duplicate, missing, or unexpected keys and reports failures separately.

Two reproducibility layers. First, local checks verify frozen receipt schemas, assignment accounting, source/input hash bindings, encoded transition semantics, and replay under specified contracts. They require neither new model responses nor a network listener. Their results apply to the artifacts and properties actually checked, not every table, binary, or platform. Bitwise identity requires support from the exact source, inputs, toolchain, and compared bytes. Matching archived hashes does not establish identical regeneration of provider responses or build outputs across machines.

Second, recollecting v5 or native-model episodes requires a separate experiment with model-admission and response-identity evidence, authorized access, budget, deadlines, and a frozen protocol. Local receipt/replay checks cannot supply missing served-model identity or recover unretained CLI output. No provider recollection was performed for this revision. Deterministic accounting and replay therefore do not imply universal portability or bitwise reproduction of historical model collections.

The public review bundle withholds the private Gemini conversation URL and original reviewer-source bytes. Under `research/academic-pipeline/gemini-review-20261113/`, `reviewer-source.public.json` and `public-source-projection-v1.json` identify the sanitized derivative. Run `python -B scripts/verify_paper_claims.py --public-source-projection` to check that derivative, frozen results, and other original-hash commitments. This explicit mode does not independently reverify withheld original-review bytes. Default strict verification remains separate, without hidden fallback or a new experiment.

TABLE V
RETAINED EARLIER EVIDENCE LANES: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	Design	Unit / budget	Comparison	Ceiling
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 defined smoke predicates; 5 archetypes	authored fixture paths	engineering-only conformance

This table summarizes the retained earlier E1/E3/ENG1 packets. The new graph audit (416 assignments per arm) and separate surface sentinels (12 per arm) have their own denominators in Sec. VI-D. Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

V. EVIDENCE LANES AND BOUNDARIES

The offline evidence comprises the original single-world E1 packet and a separate 32-graph/state, 416-assignment-per-arm authorization-boundary audit (Secs. V-B and VI-D). Their cohorts and denominators remain separate; the later audit does not convert E1 into a multi-world result.

A. *E1 Offline Conformance Design*

PQ1–PQ5 test immutability, repair, checksum/semantic/linkage/continuity faults, rollback, fail-closed assignment/adapter/manifest accounting, and unknown-key rejection. The packet contains 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures, with bindings for assignments, schemas, results, and code. Latency/token constants test accounting, not population or detector performance. E1 assigns $K{=}0$ to `rejection_only` and $K{=}1$ to unchanged retry, guided repair, and the state-reading oracle. Only the three repair-enabled arms share a budget; rejection-only is the no-repair control, not a matched-$K{=}1$ condition.

B. *Deterministic Authorization-Boundary Audit*

The frozen `research/academic-pipeline/gemini-review-20261113/protocol-v2.json` defines the designer-authored offline audit separately from E1, NPC-DISC-1, v5, and the prospective pilot. It crosses two room counts (4, 8), two topologies (locked chain, unlocked bypass branch), four start/goal placements of two keys, and two quest stages: $2 \times 2 \times 4 \times 2 = 32$ graph/state combinations. Thirteen candidate variants per combination yield 416 assignments per arm. Variants include permitted signal/observe actions and targeted policy, knowledge, disclosure, stage, object, and precondition violations. These counts specify the design, not observed successes or stochastic replicates. The 1,664 main assignments cross these authored states, variants, and arms; they are not independent samples from a game or model population. The 48 surface assignments have a separate denominator.

Graph and reachability. Typed records specify rooms, a start room, bidirectional edges with required-key sets, and object-to-room placements. Each chain has two keyed boundaries; an unlocked bypass routes around the first. A monotone closure supplies production `WorldState.reachable_locations`; a separately implemented standard-library breadth-first search examines (room, collected reusable keys). Both assume finite undirected graphs with persistent, reusable keys collected on visits. The profile excludes key consumption, random transitions, dynamic edges, inventory limits, and irreversible branches. Entity names and generated prose cannot supply missing edges or placements. Agreement checks reachability within this profile, not full-game playability.

Interventions and reference. The `schema_only` arm ($K{=}0$) uses the strict shared parser, bypasses semantic gating, and applies parsed effects/stage in an isolated simulator. This is an executed intervention, unlike E1's derived schema-only counterfactual. `semantic_rejection` uses $K{=}0$; `unchanged_retry` and `guided_repair` use $K{=}1$. The reference extracts the existing ordinary-rule predicate and implements BFS without importing the production validator, runtime, repair operator, or closure. It evaluates initial and repaired final candidates against the same immutable *pre-action* state, differing from the production state only in BFS-derived reachability. Goal evaluation uses the resulting state. This implementation separation reduces gate-as-evaluator circularity; it is not independently authored or blinded ground truth.

Funnel and retention. Pre-gate invalidity divides reference-invalid initial candidates by all assignments. Repair recovery divides initially invalid cases ending in a reference-valid commit by all initially invalid cases, including in the $K{=}0$ controls. Holds and failures each use all assignments and are reported separately. Unsafe commits use both assigned and committed denominators. False rejection divides initially reference-valid cases ending in hold by all initially reference-valid cases. Goals use all assignments. Exact structural retention divides distinct initially valid signatures preserved unchanged in a reference-valid commit by all distinct initially valid signatures; case-level preservation is also reported. Signatures include action type, preconditions/effects, required objects, used/disclosed facts, and both quest-stage fields, but exclude IDs, prose, metadata, locations, and graph labels. Distinct initially invalid and valid repaired signatures are counted separately. These measures assess encoded structure, not lexical diversity, creativity, or player-rated quality.

Surface sentinels and reproducibility. Six authored secret paraphrases (three English, three Korean) are paired with empty versus explicit forbidden `disclosed_facts` declarations at stage 0, yielding a separate 12-case denominator per arm. The pairs test declared-field coverage and whether prose directly changes state, not model-generated leakage prevalence. Exact inputs, source/input hashes, and the assignment inventory are frozen before outcomes. Each assignment records commit, hold, or failure, attempts, state hashes, replay evidence where produced, and measured local wall-clock time. The audit includes no model calls, model latency, or human efficacy observations.

C. *Archived live-screening materials*

Archived live-screening receipts provide immutable provenance and hash evidence only. They are excluded from active tables, counts, and claims and are never transferred to `gpt-6.1-sol`.

D. *NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions*

The stable v5 NPC collection contains 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$). They cover bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline cases, and replay. P uses authenticated Codex OAuth without fallback. O is 100% auxiliary-source, not independent gold. The verified-run-006 records preserve status, counts, latency, and reproduction information; verified-run-002/003/004 are historical. ENG-X01/SCR-X01, D02/QA-001, C-RESULT, Frozen8/Frozen16, v1–v4, and `native012` provide provenance, not C6 evidence.

E. *E3 KG/Ontology Link-Proposal Simulation*

E3 tests six authored graph-edge-withheld targets against seven fixed selection/summarization strategies in a closed-world simulation. It is engineering-only and non-authoritative, not an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction,

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

Strategy	Precision	Recall	MRR	Brier	Sem@3
S0 degree baseline	0.000	0.000	0.282	0.315	0.556
S2 typed-lexical	1.000	1.000	0.944	0.131	1.000

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	**Info source**	**Commits / cases**			**Mean edits**[e]
		G-rep.[f]	**O-only**[f]	**Irrep.**[f]	**Fields**
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	**Exact outcome**	**Interpretation**
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The table reports simulation construction results, not OWL/SHACL conformance, runtime retrieval, semantic completeness, or efficacy.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The loader admits only one valid control plus one isolated instance of each implemented validator code; therefore 13/13 agreement and all 12 implemented codes are *harness construction invariants*, not measured success. Beyond this setup, 12 negative fixtures matched authored expected codes rather than another code, and every noncommit path preserved complete canonical state. This is single-world-state structured-field-oracle conformance, not independent semantic-label accuracy.

B. Repair-Arm Comparison on Designed Fixtures

Frozen classes for 12 initially invalid designed cases are 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. ρ committed 5/5, 0/1, and 0/6, respectively, editing a mean of 1.0 fields per commit. It reads only the prior candidate and typed validator errors E, its sole state-derived input; failed arms preserve canonical state. The state-reading reference callback committed 5/5, 1/1, and 0/6, respectively. Blind never converts errors, ρ handles only payload-sufficient cases, and the oracle adds state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. Frozen-fixture operator conformance, not live-model repair quality.

C. Integrity, Boundaries, and Assignment Accounting

All 10 prespecified detectable faults were rejected by designated checks (10/10); without stable typed detector codes, no detector-layer attribution follows. A known provenance-boundary fixture changed the invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay (1/1 undetected): replay neither authenticates repair generation nor blocks writers recomputing unkeyed checksums. Two open sentinels—unextracted narrative disclosure and simultaneous candidate/policy omission of a required object—were accepted at the encoded layer (2/2), with 0 safety passes. The former unknown-top-level acceptance became a closed regression: proposal and replay parsers rejected a complete 12-field candidate with one unknown key (1/1), showing parser rejection parity, not semantic safety. Seven adapter/accounting assignments yielded 1 commit, 1 symbolic fallback, and 5 adapter failures; synthetic telemetry was populated and propagated for 2/7 (schema-pinned accounting, not measurement). All 3/3 injected duplicate/missing-assignment guards failed closed; these are raw counts from synthetic offline fixtures.

D. Deterministic Audit Results and Interpretation

Each arm received the same 260/416 reference-invalid initial proposals, across 1,664 main assignments (416 per arm). The schema-only arm committed all 260 as reference-invalid final candidates; each of the three semantic-gate arms had 0/416 reference-invalid final commits. Guided repair recovered 128/260 initially invalid proposals into reference-valid commits, held 132/416 assignments, and reached the objective in 252/416 assignments. The repaired commits distinguish recovery from rejection of every initially invalid case.

All arms preserved exact signatures for 156/156 initially valid cases, retaining 11/11 distinct valid signatures with 0/156

false rejections. These are structural, not creativity or linguistic-diversity, measures. Closure/BFS agreement was 32/32. In the separate surface sentinels, each semantic-gate arm admitted all six undeclared authored paraphrases and held all six explicit forbidden-fact declarations. This contrast identifies a declared-field boundary, not natural-language leakage prevalence or renderer safety. The counts retain separate outcome denominators and concern an authored finite-graph audit with an implementation-separated shared-rule reference, not language-model or human efficacy.

Reachability implementation crosscheck: 32/32. Surface sentences are designer-authored sentinels, not natural-language leakage prevalence or renderer safety evidence. Model and human efficacy remain unmeasured.

E. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira delivers the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent/public observations; players acquire/install, while the NPC commits only the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await; results/new observations choose the next plan, not single-candidate validator-error repair.

NPC-ENG1 has 16 deterministic R-arm engineering cases from actual engine snapshots (assigned_failure=2, completed=14); NPC-SCR1 retains all 72 P/R/O assignments and failures, 24 per arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures—not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter is the configured selector, not a provider revision; provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table XII gives bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates; efficacy pending, tokens/cost unmeasured. A read-only CLI and isolated working directory were used; tool-free inference and adversarial confidentiality were not established. Immutable verified-run-006 is revalidated against archived source bytes dated 2026-10-17 (revision fa9fb8a); this historical packet result does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. Assigned failures remain in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception allows only an adjacent leading observation request/decision pair in a positive-epoch snapshot; epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

F. Frozen `reveal_hint` contract and disclosure evidence

Only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four result files are consumed. Packet SHA-256 `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028` was checked against its sidecar. The declared normal stratum has 48 cases under `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary 4 and malformed 8 are excluded. Trace expectations match packet case IDs, not gate output.

Content-first explanation maps typed input (operation, actor, fact, stage, knowledge, prior disclosure) through ontology entities/relations and authored rules to checks, an allowed/denied decision, and canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once; denial/redisplay preserves canonical state. Explanatory only: no hidden facts, natural-language implications, or live capture.

Runtime declaration: `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, provenance `frozen_expectation_packet`. Scope contains only four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and eligible actor `captain_mira`. Rule IDs: DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; approved templates only `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

Python evaluator materializes the same cases/expectation object in three isolated arms, recording semantic outcomes, check traces, errors, before/after state SHA-256, rendered template IDs, and replay equality. Godot separately owns the existing `reveal_hint` writer and canonical mutation at the game-execution boundary; this producer neither reconstructs nor replaces Godot state. Traceability binds operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source set. 3 source hash(es) differ from the frozen baseline (game controller, lighthouse state machine, ontology-signals interview source as a non-identical public derivative). The packet hash is preserved; source-specific hashes and reproduction information remain in the authors' producer record, which is not deposited. The public derivative is not byte-equivalent to the private historical source; it is publication-materialization provenance only and implies no new gameplay or model observation.

Table XIV reports the three arms over the same 60 cases with separate normal/boundary/malformed denominators; unchecked violations are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes

Arm	Assigned	Initial invalid	Recovered	Hold	Unsafe commit	False rejection	Goal	Failed
schema_only	416	260/416	0/260	0/416	260/416	0/156	352/416	0
semantic_rejection	416	260/416	0/260	260/416	0/416	0/156	124/416	0
unchanged_retry	416	260/416	0/260	260/416	0/416	0/156	124/416	0
guided_repair	416	260/416	128/260	132/416	0/416	0/156	252/416	0

TABLE IX
DETERMINISTIC OFFLINE BOUNDARY AUDIT. RECOVERY USES INITIALLY INVALID CASES; FALSE REJECTION USES INITIALLY VALID CASES. GOAL ATTAINMENT IS SEPARATE FROM RULE VALIDITY.

Arm	Valid signatures retained	Cases retained	Invalid signatures	Repaired signatures	Unsafe/commits
schema_only	11/11	156/156	17	0	260/416
semantic_rejection	11/11	156/156	17	0	0/156
unchanged_retry	11/11	156/156	17	0	0/156
guided_repair	11/11	156/156	17	2	0/284

TABLE X
EXACT ENCODED SIGNATURES ONLY; NEITHER CREATIVITY NOR LINGUISTIC DIVERSITY IS MEASURED.

Arm	Separate surface assignments	Undeclared commits	Declared holds	Failed
schema_only	12	6/6	0/6	0
semantic_rejection	12	6/6	6/6	0
unchanged_retry	12	6/6	6/6	0
guided_repair	12	6/6	6/6	0

TABLE XI
DESIGNER-AUTHORED SURFACE SENTINELS; SEPARATE ASSIGNMENTS, NOT NATURAL-LANGUAGE LEAKAGE PREVALENCE OR RENDERER SAFETY EVIDENCE.

TABLE XII
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	**Assigned**	**Complete/fail**	**Goal reached**	**Recovery E/U**
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

TABLE XIII
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	**Value**
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This is an authorization-conformance limit under the shared frozen rules, not general ontology equivalence or ontology superiority. Figure 4 distinguishes E/R (expected-match/replay cases), C (check-record PASS/FAIL), and NR (cases with no reported checks); NR is not skipped, pass, or out-of-scope. Table XV maps rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is unchanged.

Claims are limited to bounded engineering conformance: no human gold labels, access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. The authors' producer record binds input identities and hashes; it is held locally and no reproduction commands are provided here.

ENG1. Legacy Godot 4.7.1 replays authored rejection/commit against prior/offline hashes. Presentation-only captures show the board, hints, coach, chain prompt, and dashboard checks. The retained `game-track/godot/docs/latest/evaluation-matrix.json` reports four fixtures, 52/52 core checks, and 4/4 extension checks separately. These stored engine-local observations are not a fresh rerun or evidence of transport, participants, narrative, usability, or efficacy. The smoke contract defines eight named predicates, not a verified historical smoke-run pass.

Figure 5 shows four authored native screenshot stages of the fixed game's proposal/authorization interface, not a continuous observed playthrough. The captures do not recollect historical model responses or add model, participant, or prospective-pilot assignments.

TABLE XIV
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XV
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

G. Single-task Codex development observation

The frozen `ontology-map-20261023-01` record covers one read-only bilingual dashboard task (n=1, independent repeats 0), not an NPC trial/cohort. Six native CLI invocations (initial + five resumes; one read-only/no-edit) belong to one case. `turn_context.model=gpt-6.1-sol` records runtime selection, not provider attestation. Provider identity/revision/seed/cost are `null`; model-only latency is unavailable. Wall-clock 5942.665 s and CLI 263.32 s measure neither inference latency nor the future endpoint.

Invocation six repaired the inherited 660 px mobile-table minimum width after geometry RED (2 failed/35 passed), followed by 62/62 map/signal/Web checks, including cell/paragraph geometry, and a successful Godot Web build. Browser attempt 10 passed; 01–08 failed setup/routes; 09 passed older gates despite clipping.

The final $(\text{seq}, t_{\text{ms}})$ observations were stage-0 denial $(68, 19468)$ with equal hashes, fresh disclosure $(239, 58234)$ with different hashes, and authorized redisplay $(249, 59793)$ with equal hashes. At 1440×900/390×900, the driver observed Enter, inspection/frozen-comparison invariance, and zero pre-navigation console/page/network errors/warnings.

A later production-dashboard rerun is recorded separately as **NOT_PASS**: the headed run stopped at the unchanged yaw guard; the headless run passed seven component gates but logged four pointer-lock page errors. This does not change attempt-10 PASS under its dated acceptance set. The case demonstrates neither full-game creation, a Python–Godot authorization round trip, nor a model outcome. Privacy/source-binding failures and same-thread repairs add no n. Jev remains `proposal_only`, without label/approval authority. The dated 62/62 result comes from `research/development-cases/ontology-map-20261023-01/evidence/verification-009-mobile-complete-tests.json`. That command receipt identifies the three test files and exit code 0; its referenced stdout records 62 passed tests. This is an archived command observation, not a fresh execution or a count inferred from current test definitions.

The native eight-assignment pilot tests the fixed quest's provider interaction, validation, engine commit, and replay pipeline. Its authored goal/failure cases establish neither selector superiority, efficacy, nor statistical generalization.

H. Fixed-game native runtime comparison

This public packet retains 8 assigned rows for one authored quest (`sealed-lighthouse-v1`), with seed 20261024 and two repeats per selector/representation cell. System-endpoint successes are 5/8; observed native executions are 7. Host invocations are summed from actual receipts (23), and observed repairs sum to 0. Committed transitions total 16, of which unsafe are 0; semantic replay valid/invalid/NR is 7/0/1. Fresh-engine reexecution disposition counts are VALID=6; INVALID=0; NOT_RUN=1; NOT_OBSERVED=1. Observed elapsed time is n=7; 48.19/41.09 seconds (mean/median), and preparation time is n=7; 6.67/6.30 seconds. Unreported values are NR, not zero.

The first row is retained as a harness JSON-serialization technical failure and is not attributed to GPT-6.1 capability. At least 2 GPT-6.1 ontology row(s) retain a response-timeout failure under the frozen 60-second bound. Models provide typed-intent proposals; the deterministic Python planner/checker and Godot commit retain authority. The table is therefore a technical observation for one quest, not an efficacy, superiority, noninferiority, general ontology-consistency, or player-benefit claim. Semantic record-link replay is not fresh-engine reexecution. Provider identity/revision/cost and provider-request counts remain NR outside the public observation boundary.

I. Separate Synthetic Development Collection (v5) and Two-Stage Decomposition

Archived cohort scope. Earlier v1–v4 and diagnostic `codex-feedback-pilot-v2-*` cohorts remain immutable history outside active v5 denominators. They were not all aborted: v4 Luna completed all ten assignments. `research/codex_feedback_pilot/public-history`

TABLE XVI
FIXED-GAME NATIVE RUNTIME COMPARISON (N=2 PER CELL). ALL EIGHT ASSIGNMENTS, INCLUDING THE HARNESS FAILURE AND TIMEOUTS, REMAIN IN DENOMINATORS.

Requested selector	Form	Goal/assigned	Observed exec.	Host calls	Unsafe/committed	Elapsed median (s; n)	Fresh engine
gpt-6.1-sol	general_rule	1/2	1	4	0/3	45.91 (1)	1/0/0/1
gpt-6.1-sol	ontology	0/2	2	3	0/1	72.35 (2)	1/0/1/0
gpt-5.6-sol	general_rule	2/2	2	8	0/6	40.24 (2)	2/0/0/0
gpt-5.6-sol	ontology	2/2	2	8	0/6	33.11 (2)	2/0/0/0

Fresh engine counts follow VALID/INVALID/NOT_RUN/NOT_OBSERVED order. NR is unreported, not zero. Overall counts and the separate semantic replay are reported in the text.

`-20261111/cohort-inventory.json` preserves assignment counts and raw-source digests for 19 entries: 17 archived cohorts and two v5 selector collections. The separately attributed exploratory signal example is outside this inventory and active denominators. Its original receipt is unavailable, so its historical attribution is not source-verified (Appendix D).

The separate v5 collection uses five co-designed synthetic development fixtures, not a held-out test. Each selector received Detailed and Generic feedback for each fixture (10 assignments each), in nominal serial order Luna then Terra. All 20 assignments received responses; receipt does not imply task success. Table XVII reports gate acceptance, the independently specified goal, and schema failure over all five assigned cases per cell (C4), including failed outputs. These 20 fixed fixture/feedback assignments test the authored response–parse–authorize–commit–goal pipeline. They are descriptive conformance cases, not independent samples establishing superiority, efficacy, or statistical generalization.

Stage 1—syntactic parser (C1). Strict parsing rejected 10/20 responses (Luna 1/10; Terra 9/10), leaving ten parsed responses. The frozen evaluator parses before controller execution, so schema failures precede semantic validation and commit. Their `CandidateParseError` records have null evaluations: they do not contain observed transitions, independently measured zero-mutation outcomes, or evidence of content-level failure causes. These rejection records establish a syntactic check outcome only. These ten unevaluated assignments receive no semantic-safety credit.

Stage 2—seven-predicate semantic gate (C2). Among the ten parsed, identity-preserving responses, the gate accepted 8/10: Luna Detailed 4/4, Luna Generic 3/5, and Terra Generic 1/1. Terra Detailed had no parsed response. The two rejected parsed responses fell back with state unchanged (zero mutations). Independently, 7/20 assigned responses reached `signal_restored`. The collector made no further repair calls (`repair_budget=0`); failures remain in every assigned denominator (C4).

E1's 12 semantic-negative fixtures satisfied the candidate contract, so a schema-only boundary would reject 0/12. This remains a derived counterfactual, not an executed E1 arm. The schema-only intervention in Sec. V-B has separate assignments and outcomes from E1 and v5.

Table XVIII partitions the same five fixtures in a *post-hoc descriptive* analysis: four permit progress toward `signal_restored`; `unauthorized_objective` is the preexisting policy–goal-conflict negative control. This is not a preregistered endpoint or new cohort, and Table XVII's five-case denominators remain unchanged. Authorized-progress goals are 3/4 for both Luna conditions, 0/4 for Terra Detailed, and 1/4 for Terra Generic, with parser failures counted as assigned nonsuccesses. In the conflict case, Luna Detailed suppressed the forbidden goal through a policy-compliant no-effect `COMMIT`; Luna Generic used `HOLD`/fallback. Terra's two schema failures are unassessed, not safe holds. Suppression is confirmed for 2/2 observed conflict outcomes; 2/4 assigned outcomes remain unassessed, not 4/4 safety. Luna Detailed's 4/5 gate acceptance and 3/5 goal completion measure different predicates, not a goal advantage.

The five fixtures are `missing_precondition`, `missing_effect`, `unauthorized_objective`, `forbidden_disclosure`, and `unsatisfied_dependency`. The frozen amendment permits continuation after schema-invalid output only when adapter completion is proven; genuine transport or safety failures still stop collection. In `unauthorized_objective`, the frozen `signal` policy permits no effects, but the goal requires `signal_restored`. One Luna Detailed response removed the forbidden effect and committed a policy-valid candidate without reaching the goal. Table XVII's 4/5 gate and 3/5 goal counts aggregate five fixtures, not this conflict case alone. Detailed and Generic feedback carry unequal information. Served model identity/revision, request count, seed, and cost remain unknown. This small co-designed collection establishes neither selector superiority nor ontology efficacy and confirms neither A nor C. The conflict is part of the frozen fixture: the objective requires an effect its policy forbids. The accepted no-effect result is a safe `COMMIT`, not a `HOLD`; empty effects do not change its recorded terminal class. Its goal failure is a negative-control outcome, not failed authorized recovery.

The authors hold the public-safe `research/codex_feedback_pilot/public-v5/` projections pending deposit. They preserve case outcomes, prompt hashes, and executable hash/version; they omit the local executable path and disclose the original-file digest and omitted field. Frozen raw outcomes and approvals remain local and unchanged. The immutable `research/codex_feedback_pilot/paired-model-comparison-20261110-v5-source.zip` contains seven executed source files as provenance, not a standalone runnable distribution. Parsed response text is retained. Raw CLI stdout/stderr were not retained because they could expose credentials or sensitive server data; byte counts, digests, and event summaries remain. Receipt and parsing failure establish

TABLE XVII
THE SEPARATE V5 COLLECTION RETAINS ALL FIVE ASSIGNED CASES PER DENOMINATOR, INCLUDING SCHEMA FAILURES. RECEIPT DOES NOT IMPLY TASK SUCCESS.

Requested selector	Signal	Received	Strict gate accepted	Goal	Schema failure
`gpt-5.6-luna`	Detailed	5/5	4/5	3/5	1/5
`gpt-5.6-luna`	Generic	5/5	3/5	3/5	0/5
`gpt-5.6-terra`	Detailed	5/5	0/5	0/5	5/5
`gpt-5.6-terra`	Generic	5/5	1/5	1/5	4/5

TABLE XVIII
POST-HOC DESCRIPTIVE PARTITION OF THE UNCHANGED FIVE-FIXTURE V5 ASSIGNMENTS; NOT PREREGISTERED, NOT AN INDEPENDENT COHORT, AND NOT A REPLACEMENT ENDPOINT. TERRA CONTROLS ARE UNASSESSED AFTER PARSER FAILURE.

Requested selector	Signal	Original goal /5	Authorized goal /4	Control handling /1
gpt-5.6-luna	detailed	3/5	3/4	forbidden goal suppressed 1/1; COMMIT (no effects; state ID changed)
gpt-5.6-luna	generic	3/5	3/4	forbidden goal suppressed 1/1; HOLD (fallback; state unchanged)
gpt-5.6-terra	detailed	0/5	0/4	unassessed 1/1 (schema unknown)
gpt-5.6-terra	generic	1/5	1/4	unassessed 1/1 (schema unknown)

neither provider identity nor inherent model inability.

Figure 6 shows the feedback signal: Detailed carries a failed check and repair hint; Generic carries rejection alone. Neither signal is an affect estimate or authorization. Both conditions use the same parser, identity check, and seven predicates. Figure 7 tracks receipt, parsing, gate acceptance, and the independent goal for every assignment, including the authorized commit that missed the goal.

The prospective two-track protocol is `NOT_RUN/BLOCKED`, with 0/180 units executed (Appendix D); it is not a completed confirmatory pilot.

VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

The guided semantic repair operator ρ [48], [55], [56], [57] uses deterministic pre-execution errors; the oracle-only arm reads state. Rule agreement and edit-scope diagnostics establish neither model capability nor ontology superiority. Controller versions confound the native comparison: row 1 failed under controller-v3 before a response, whereas rows 2–8 used controller-v4 after a one-line JSON-boundary correction. Model-visible inputs were unchanged, but selector/representation differences cannot be attributed to the model.

The fixed-game pilot provides bounded native evidence, not ontology efficacy. A (model/stack consistency) and C (fresh-reference noninferiority) remain unexecuted. They require budget-bounded completion under fixed source/policy, sealed assignments, exact identity/revision fields, and separate safety/replay accounting; the pilot answers neither. The proposed endpoint runs from request/evidence receipt to the first receipt passing frozen mandatory checks (time-to-first-all-predeclared-validation-pass). Unfinished, timeout, failure, and stage-not-reached runs have no successful time. The proposed endpoint is measured in continuous task-clock seconds.

The identity-required extension stopped at missing served-model evidence (Appendix B); no new identity-qualified comparison was obtained.

The deterministic audit covers more authored graph/states, not more sampled models or players. Reusable-key reachability does not validate omitted mechanics; a shared rule specification can cause shared reference/production errors. Exact structural retention measures neither linguistic variety nor voice stability. Authored paraphrase pairs test a declared-field boundary, not prevalence or unrestricted-dialogue safety. No new admitted multi-model or human efficacy collection was performed. The v5 collection's 20 assignments, the native pilot's eight assignments, and NPC-DISC-1's 60 authored cases measure distinct constructs with distinct denominators. Pooling them with the 1,664 deterministic case assignments or 48 surface assignments would not create a population sample. All five C-RESULT outcomes remain `TODO-RESULT`; TR01–TR06 remain unmeasured, and prospective execution remains 0/180.

Threats and boundaries. The native pilot has one authored quest and two repeats per cell, without comparator power, neutral complete-attempt inventories, or independent traceability scores. State consistency does not establish behavioral consistency [58] or narrative coherence. Predicate/role/Godot/DISC sets are not latent states; declarations are not human gold or holdouts; evidence lanes have separate denominators. The data support no participant, affect, comprehension, agency, enjoyment, immersion, benefit, population, safety-rate, model-ranking, or commercial-performance inference. Hashes do not authenticate, and replay equality is not semantic correctness. Future human/affect studies require ethics review, consent, minimization, retention/deletion, and opt-out. Prior evaluation work motivates compensation disclosure and calibrated judges [39], [40], [41], [42].

VIII. CONCLUSION

TRACE-RPG integrates C1–C5 authorization, repair, replay, and assignment accounting with a source-bound bilingual observer. The Codex case and native pilot retain failures and evidence limits; they do not establish model performance. Whole-game, time-to-first-all-predeclared-validation-pass, traceability,

confirmatory A/C, and repeated-cross-model evaluation remain future work. Historical cohorts remain unpooled.

Disclosure of AI Usage

LLMs assisted prose, code, tests, and citations; hosted candidates were labelled. Curated AI art and the desk mesh retain file provenance; snapshots, hashes, and ledger views are presentation-only. The authors checked sources and accept responsibility. NPC inputs/tables preserve assignment/protocol/schema hashes and unknown token/cost/model-revision fields.

EXTERNAL TRACE COUNTS

	N normal	B boundary	M malformed
U	E 0/48 n=48 R 48/0 C 0/0 NR 48	E 0/4 n=4 R 4/0 C 0/0 NR 4	E 0/8 n=8 R 8/0 C 0/0 NR 8
G	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0
O	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 counts by arm/stratum. E/R: expected-match/replay; C: policy-check PASS/FAIL; NR: no reported check, not skipped. Packet-bound engineering counts measure neither latent state, free-language leakage, ontology superiority, nor player benefit.

Data and Code Availability

The authors hold the implementation/build source, frozen packets, Godot slice, deterministic scripts, machine-checked contribution, reference-topic, and experiment matrices, development-case record, disclosure/NPC artifacts, and public-safe native comparison projection pending external DOI/repository deposit. Appendix E (Table XX) gives canonical paths, schemas, and verification entry points for each lane. The traceability supplement and source manifest contain the proposed rubric, codebook, and whole-game/repeated-cross-model plan. Historical receipts and identity-admission outcomes remain unpooled.

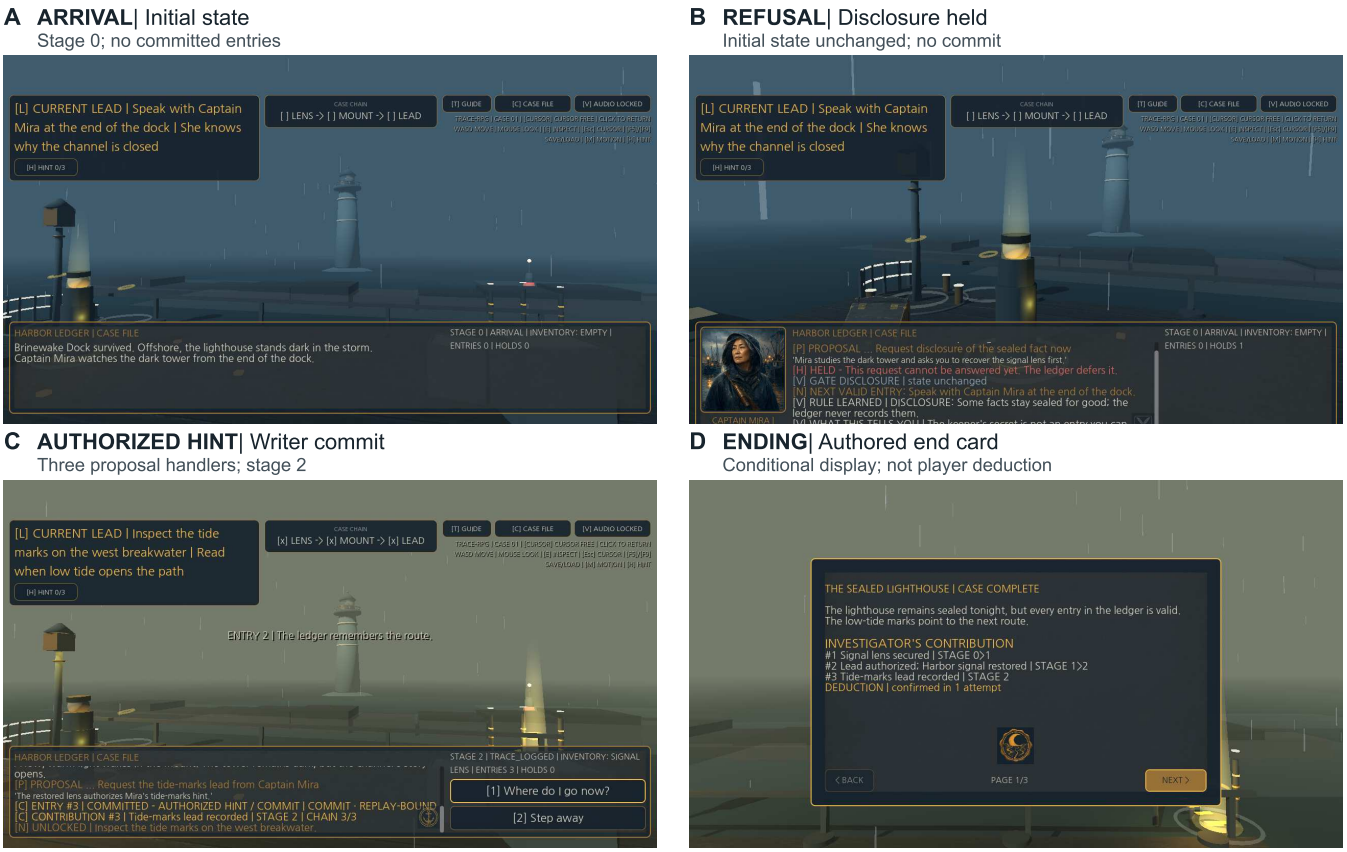


Fig. 5. Native Godot 4.7.1 engineering captures: (A) initial arrival; (B) denied secret disclosure with initial state unchanged; (C) authorized hint at stage 2 after three existing proposal-handler commits; (D) an authored conditional closing card at C's terminal state. B is not exact replay of the archived post-acquisition `evt-002` event; D is not observed player deduction. Existing curated AI portrait/seal art and lower-HUD clipping in B/C remain visible. The captures document rendered state correspondence, not efficacy, usability, or execution of the blocked pilot.

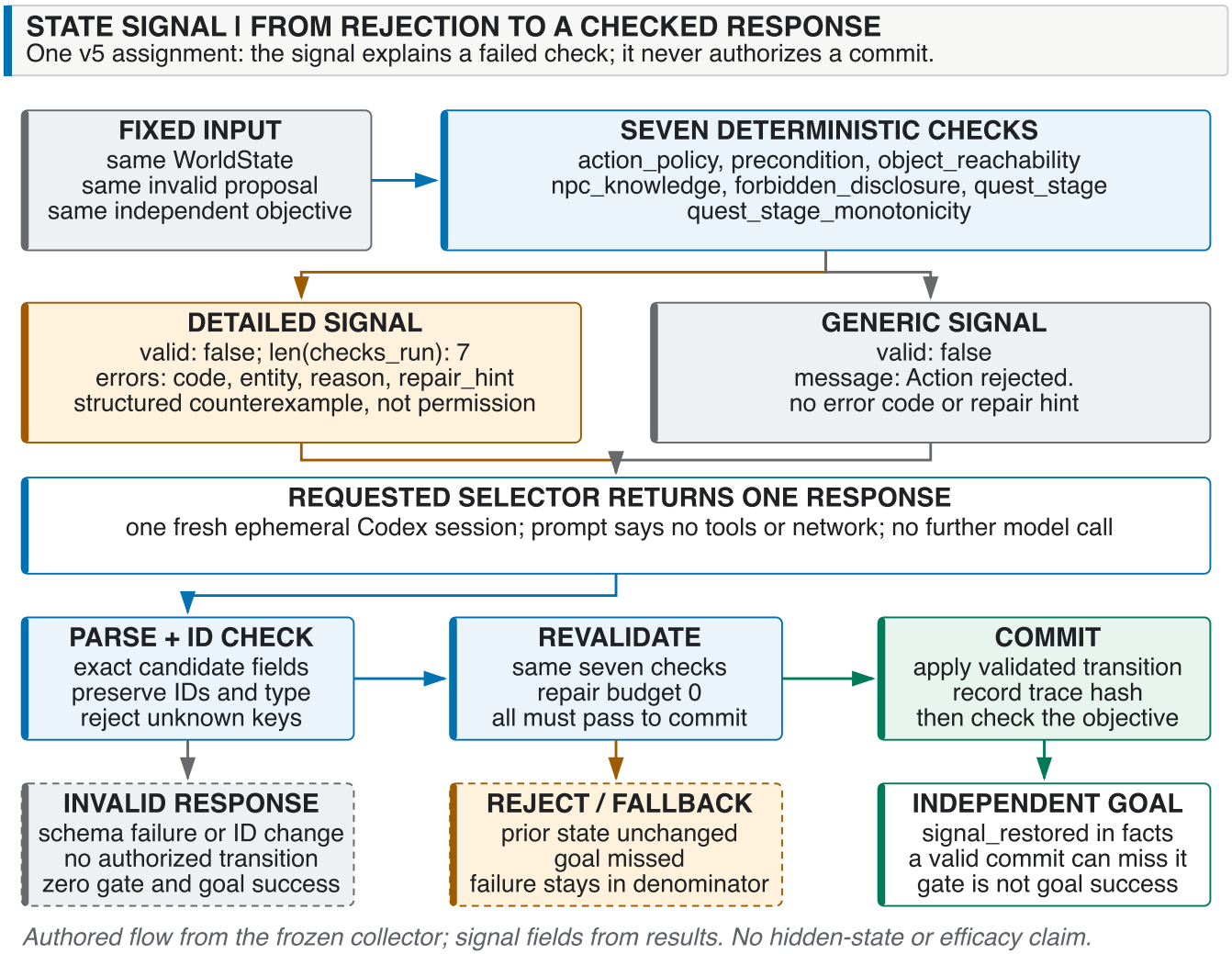


Fig. 6. In the v5 state-signal path, (ok, E) supplies repair information; revalidation alone authorizes mutation. Generic feedback omits error details. Goal evaluation is separate from gate acceptance.

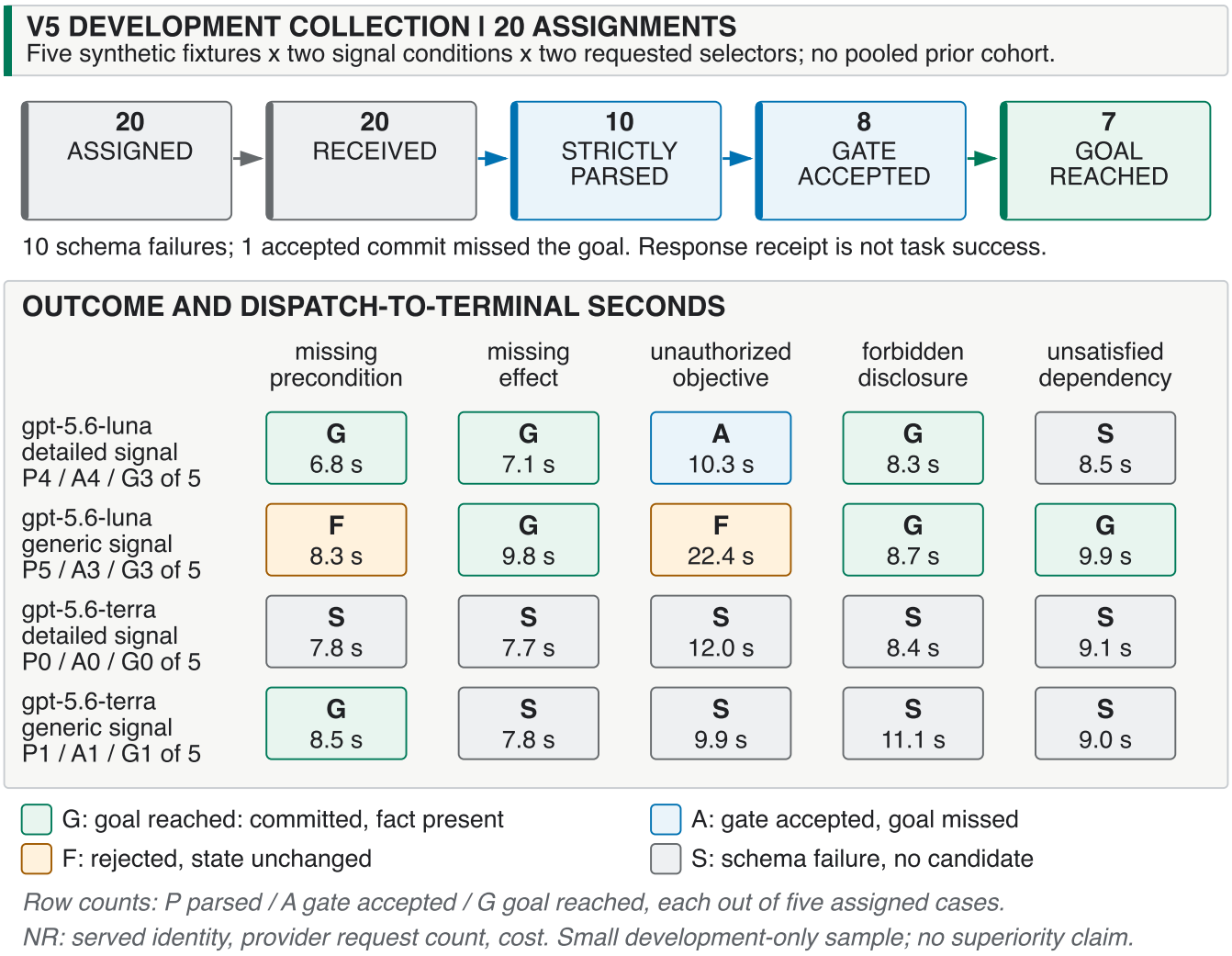


Fig. 7. All 20 v5 assignment outcomes from Table XVII's receipts. Ten schema failures remain in the denominator; eight accepted candidates yield seven goals. Seconds cover dispatch to terminal, including failed responses, not model-only latency.

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APPENDIX A
RUNTIME EVIDENCE APPENDIX

Figure 8 reports all eight assigned native-pilot episodes on `sealed-lighthouse-v1` (seed `20261024`, 2 selectors × 2 representations × 2 repeats) from `research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json` (Table XX). Each row distinguishes three verification layers: host-call receipt accounting, semantic record-link replay (`R`), and fresh Godot reexecution (`E`), alongside Table XVI:

- **Row 1 (`gpt-6.1-sol`, `general_rule`, r1 — `HARNESS`):** Controller-v3 failed at JSON-boundary serialization before any response (`executed=null`, `host_invocations=0`, `Goal=NR`, `Safety=NOT_OBSERVED`, `R NR / E NOT_OBSERVED`). The row remains in the assigned denominator (1/2 goal rate); its pre-response harness failure is not evidence of model capability.
- **Row 2 (`gpt-6.1-sol`, `general_rule`, r2 — `SUCCESS`):** Under controller-v4's one-line JSON-boundary correction, model-visible inputs were unchanged. The episode reached the goal in 4 host calls (`45.91 s`) through 3 safe commits (`0` repairs, `0` unsafe), with valid semantic replay (`R YES`) and fresh-engine reexecution (`E VALID`).
- **Rows 3–4 (`gpt-6.1-sol`, `ontology`, r1 & r2 — `TIMEOUT`):** Both episodes exceeded the frozen 60 s response deadline (`78.18 s` and `66.52 s`; `Goal=NO`). Row 3 committed 1 valid state change before a subsequent timeout; that transition passed safety (`PASS`), semantic replay (`R YES`), and fresh Godot reexecution (`E VALID`) despite episode goal failure. Row 4 timed out before any commit (`0` commits, `NO_COMMIT_OBSERVED`). Semantic replay checked its unchanged state (`R YES`); no commit triggered fresh-engine reexecution (`E NOT_RUN`).
- **Rows 5–8 (`gpt-5.6-sol`, `general_rule` & `ontology`, r1 & r2 — `SUCCESS`):** All four assignments reached the goal (2/2 per cell; 4 host calls and 3 safe commits each, `31.95--41.09 s`, `0` repairs). Each passed safety (`PASS`), semantic replay (`R YES`), and fresh-engine reexecution (`E VALID`).

FIXED-GAME NATIVE RUNTIME COMPARISON | 8 ASSIGNED ROWS

One authored quest; n=2 per selector/representation cell; NR = not reported, not zero. Labels remain explicit; color is supplementary.

Row	Requested selector	Representation / repeat	Status	Goal	Host calls	Elapsed (s)	Safety R record-link / E engine
1	gpt-6.1-sol	general_rule / r1	HARNESS	NR	0	NR	NOT_OBSERVED R NR / E NOT_OBSERVED
2	gpt-6.1-sol	general_rule / r2	SUCCESS	YES	4	45.91	PASS R YES / E VALID
3	gpt-6.1-sol	ontology / r1	TIMEOUT	NO	2	84.66	PASS R YES / E VALID
4	gpt-6.1-sol	ontology / r2	TIMEOUT	NR	1	60.04	NO_COMMIT_OBSERVED R YES / E NOT_RUN
5	gpt-5.6-sol	general_rule / r1	SUCCESS	YES	4	41.09	PASS R YES / E VALID
6	gpt-5.6-sol	general_rule / r2	SUCCESS	YES	4	39.39	PASS R YES / E VALID
7	gpt-5.6-sol	ontology / r1	SUCCESS	YES	4	36.68	PASS R YES / E VALID
8	gpt-5.6-sol	ontology / r2	SUCCESS	YES	4	29.55	PASS R YES / E VALID

Endpoint outcomes, receipt-derived host calls, and dispatch-to-terminal elapsed time.

Typed-intent proposal; deterministic planner/checker and Godot commit retain authority.

No efficacy/superiority/noninferiority/player-benefit claim; provider metadata and request counts remain NR.

Fig. 8. Native fixed-quest pilot from recorded traces: one authored quest, 8 assigned rows, 7 observed executions. Provider identity, revision, cost, and request count remain unknown. The assigned denominator retains one pre-response harness-serializer failure and two ontology response timeouts under the frozen 60 s `gpt-6.1-sol` deadline, separately from goal successes. This is a trace visualization, not a gameplay screenshot.

APPENDIX B
IDENTITY-REQUIRED EXTENSION

The admission gate in `research/runtime-model-swap-20261024/codex-identity-20261026/public-results-v1/results.json` (Table XX) requires response-bound provider model attestation (`openai-model`) before admitting an episode to a cross-selector comparison.

Identity-admission stop: not a model comparison. The separate extension has 20 planned rows, 1 assigned row, 1 native process, 0 identity-qualified rows and 19 unattempted rows. The first requested `gpt-6-luna`/`general_rule` CLI turn completed, but its response lacked provider model-identity evidence; it was not accepted and the batch stopped.

A requested selector or catalog advertisement is not a served identity. The evidence cannot distinguish an absent `openai-model` header from one not delivered as a valid string. This is not an observed model mismatch or model-incapability result. `gpt-6-terra` was not advertised by the pinned catalog, was not assigned and was not substituted.

Audit distinction: `record hash valid=true`; `engineering record accepted=false`. The original bootstrap label `episode_record_integrity_invalid` is retained. The recomputed hash matches; the compound predicate also requires engineering/safety PASS, so this label is not evidence of corrupted bytes. There were 0 commits, safety is `NO_COMMIT_OBSERVED`, and fresh-engine replay is `NOT_RUN`, not a safety pass.

Two blocked repair requests were not native reruns. The original eight-row pilot remains byte-identical and unpooled. Provider identity, revision, effective effort, request count, cost, actual credit use and cash charge remain `unknown`. Matching accounts and automatic reload OFF were observed before and after, without a setting change or purchase; these are not zero-cost or zero-credit attestations. No verified new model comparison, ontology effect or confirmatory A/C evidence follows.

Preparation 7.144 s; episode elapsed 4.257 s; native process 4.207 s; orchestration 11.566 s. These are distinct failed-attempt boundaries, not pure inference latency or time to success.

TABLE XIX
TEN IDENTITY-ADMISSION CELLS DISTINGUISH PLANS, ASSIGNMENTS, QUALIFIED IDENTITY AND UNATTEMPTED ROWS; NOT MODEL PERFORMANCE.

Requested selector	Representation	Planned	Assigned	ID-qualified	Not run
gpt-6-luna	general_rule	2	1	0	1
gpt-6-luna	ontology	2	0	0	2
gpt-6-sol	general_rule	2	0	0	2
gpt-6-sol	ontology	2	0	0	2
gpt-5.6-luna	general_rule	2	0	0	2
gpt-5.6-luna	ontology	2	0	0	2
gpt-5.6-sol	general_rule	2	0	0	2
gpt-5.6-sol	ontology	2	0	0	2
gpt-5.6-terra	general_rule	2	0	0	2
gpt-5.6-terra	ontology	2	0	0	2
Total	—	20	1	0	19

APPENDIX C
WITHDRAWN OLLAMA HISTORICAL ATTRIBUTION

The earlier local Ollama quantitative results, eligibility claims, and engineering-pass attribution are withdrawn because their original episode/response receipts could not be recovered. `research/maintenance/ollama-unverified-history-20261113.json` preserves the former section and table as unverified historical attribution withdrawn from current quantitative evidence, not original raw data or a validated experiment. No substitute observations, replacement receipts, or model recollection were produced. Missing originals remain an explicit limitation, not a passing or silently skipped verification requirement.

APPENDIX D
DIAGNOSTIC, HISTORICAL, AND BLOCKED COHORTS

A. Exploratory Signal Example

The exploratory example is an unverified historical attribution: its original receipt is unavailable and it is outside the 19-cohort inventory. It remains excluded from active denominators and confirmatory evidence. The historical account describes one synthetic `missing_effect` fixture that omitted `signal_restored`. It attributes objective success after one repair to a deterministic control. Under the requested `gpt-6-luna` selector, it reports accepted repair and objective success with Detailed feedback, but empty effects, rejection, and unchanged-state fallback with Generic feedback (`Action rejected.`). These attributed outcomes explain the intended contrast in Fig. 6; they are not independently verified observations. The available artifacts do not establish complete prompts, executed source-version bindings, or served-provider metadata.

B. Incomplete Synthetic Development Pilots and Bookkeeping

`research/codex_feedback_pilot/public-history-20261111/cohort-inventory.json` (Table XX) preserves assignment accounting and raw-file SHA-256 digests for 19 entries: 17 archived cohorts and two v5 selector collections. The earlier v2 evidence and v2 diagnostic cohorts for `gpt-5.6-luna` and `gpt-5.6-terra` (10 planned assignments each) stopped at their first `missing_precondition` assignment with `protocol_error` (`FAILED`) or remained `IN_PROGRESS`/pending with 0 valid responses. Raw CLI error text was omitted to prevent credential leakage. Completed/failed/pending counts across the 17 archived cohorts are: original pilot $0/7/3$ (seven technical/interrupted attempts); four initial selector/native files $0/0/40$; initial v2 pair $0/0/20$; startup diagnostics $0/2/18$; v3 pair $0/0/20$; v4 Luna $10/0/0$ (detailed goal $4/5$, generic goal $4/5$) and Terra $0/1/9$; plus v2 evidence ($0/1/9$ per selector) and the v2 diagnostic snapshots above. The archive is therefore not uniformly aborted. None of these 17 archived records is pooled with v5 in Table XVII.

C. Pre-Registered Fail-Closed Blocking Gate and Zero-Leakage Check

The pre-registered `research/prospective-pilot-20261103/protocol-v1.json` and `results-v1/` specify three selectors (Codex `gpt-5.6-sol`, Codex `gpt-6.1-sol`, Antigravity `gemini-3.8-flash-medium`; nominal medium effort) in two tracks (Table XX). Runtime crosses 3 targets $\times$ 2 information-matched representations $\times$ 10 fresh resets (60 units, 180 s/episode). Bounded development crosses 3 targets $\times$ 2 exploration modes (minimal input vs. repository exploration) $\times$ 2 representations $\times$ 5 tasks $\times$ 2 fresh runs (120 units, 900 s/assignment). The primary endpoint is budget-bounded validated success, with hard safety as a separate gate.

Prospective pilot status: not a model comparison. The frozen study `prospective-pilot-20261103` plans 180 units (60 runtime and 120 development) over three requested targets. 180 are blocked, 0 assigned, 0 attempted and 0 executed, with 0 native process starts. Status `NOT_RUN/BLOCKED`; scientific conclusion `NOT_MET`.

Success rate: null. Observed cost: null (known subtotal null; 0 admitted units with unknown cost). Effect estimate: null. A null is an absent measurement, never zero.

Blocked prerequisites, counted per unit: `BLOCKED_ACTIVATION_ABSENT` (180), `BLOCKED_IDENTITY_FORMAT_UNOBSERVED` (180), `BLOCKED_ISOLATION_UNPROVEN` (180), `BLOCKED_LIVE_PATH_NOT_IMPLEMENTED` (180), `BLOCKED_NO_CREDIT_ENFORCEMENT` (180), `BLOCKED_PUBLIC_DERIVED_UNAPPROVED` (180), `BLOCKED_SCHEMA_ACCEPTANCE_UNOBSERVED` (180), `BLOCKED_TARGET_UNVERIFIED` (60), `BLOCKED_TASK_FREEZE_INCOMPLETE` (180).

The earlier eight-row 5/8 pilot, the stopped identity extension, the Ollama lane, the deterministic fixtures and the n=1 results are separate lanes. They are neither pooled nor re-reported here.

Nine fail-closed admission gates blocked all 180 units before any model process launched. Subsequent independent offline controls verified acceptance of two correct variants and rejection of the defective baseline and plausible-wrong variant for all five development tasks. The corpus leakage audit passed within its declared scope over 9,303 repository files. These checks establish pre-collection gating, fixture readiness, and leakage-check conformance, not an executed pilot or model comparison. No blocked run was fabricated or retroactively admitted.

APPENDIX E
EVIDENCE ARTIFACT AND DIRECTORY REFERENCE

Table XX indexes canonical paths, recorded fields, and verification entry points for contributions (**C1–C5**), active and historical lanes (**E1–E3**, **ENG1**, **NPC-ENG1/SCR1**, **NPC-DISC-1**, **DEV-CASE**), and runtime/development cohorts.

TABLE XX
CANONICAL EVIDENCE DIRECTORIES, FILES, RECORDED FIELDS, AND VERIFICATION SCRIPTS FOR CLAIMS, LANES, AND APPENDIX COHORTS.

Lane / Scope	Directory and Key Files	Core Contents and Recorded Evidence Fields	Verifier & Reference
C1–C5 Core Contracts & ρ	`src/nesy_game/contracts.py`, `validator.py`, `repair.py`, `runtime.py`, `experiment.py`; `game-track/schemas/*.schema.json`	`WorldState`, `ActionPolicy`, 7 predicate checks (`VALIDATOR_CHECKS`), guided repair $\rho(a, E)$, state-reading oracle, semantic replay & SHA-256 chain verifier, Draft 2020-12 bridge/record schemas.	`scripts/validate_project.py` Sec. IV, Tables II–IV, Fig. 3
E1 Offline Conformance	`research/academic-pipeline/stage-04-pilot/pilot-results.json`, `pilot-assignment-manifest.json`, `sha256-manifest.json`, `*.csv`	13 gate fixtures (1 valid, 12 negative), 12 repair cases across 4 arms (0/12 reject/blind, 5/12 ρ, 6/12 oracle), 10 integrity-fault checks, 7 adapter terminal records, 3 manifest guards, 2 open sentinels.	`scripts/run_conformance_pilot.py` Sec. V, VI, Tables VII–VIII
Boundary Audit	`research/academic-pipeline/gemini-review-20261113/results-v2/`; `summary.json`, `rows.jsonl`, `surface_rows.jsonl`, `assignment_inventory.json`, `input_hashes.json`, `source_hashes.json`	1,664 main assignments and 48 separately denominated surface assignments; bounded authored finite-graph audit, frozen inventory, and input/source hashes.	`scripts/run_gemini_review_evaluation.py` `tests/test_gemini_review_publication.py` Secs. V-B, VI-D
E2 Archived Live Screening	`research/academic-pipeline/rq2-live-pilot/promotion-manifest.json`, `frozen-pilot-base/`, `signal-repair-v2/`	14 hash-pinned historical JSON/JSONL receipts (5 seeds $\times$ K=1 paired ρ vs. blind retry, 15/15 non-commit state isolation); excluded from active claims.	`scripts/validate_contribution_crosswalk.py` Sec. V (Historical only)
E3 KG/Ontology Simulation	`knowledge/ontology/trace-rpg-ontology.json`, `curated-relations.json`; `research/simulation/kg-ontology/latest/evaluation-matrix.json`	43 OKF nodes, 106 reference edges, 24 curated typed edges; 6 edge-withheld queries $\times$ 7 fixed strategies (210 scores; `S2` P=R=1.000, MRR=0.944, Sem@3=1.000).	`scripts/run_kg_ontology_simulation.py` Sec. V, Table VI
ENG1 & **DEV-CASE**	`game-track/godot/project.godot`; `game-track/godot/docs/latest/evaluation-matrix.json`, `game-track/godot/docs/latest/balance-archetypes.json`; `game-track/web/dashboard/ontology-map.json`; `research/development-cases/ontology-map-20261023-01/` (`contract.json`, `result.json`, `evidence/verification-009-mobile-complete-tests.json`)	Retained matrix: 4 fixtures, 52/52 core and separate 4/4 extension checks; 5 archetypes. Eight defined smoke predicates are a contract, not a stored run-pass result. Bilingual 8-role map; n=1 development case with dated 62/62 test receipt.	`scripts/validate_game_track.sh`; `tests/test_godot_experimental_game.py` Sec. VI–VI-G, Table III
NPC-ENG1 & **NPC-SCR1**	`configs/npc-goal-recovery.json`; `research/academic-pipeline/npc-goal-recovery/verified-run-006/aggregate.json`, `*-results.jsonl`	16 deterministic `NPC-ENG1` cases and 72 `NPC-SCR1` rows ($P/R/O$=24 each), 12 stale-control ACK rejections (`SCR-T01/T02`), epoch-local replay, and pinned `fa9fb8a` provenance hashes.	`scripts/validate_npc_goal_evidence.py` Sec. V, VI, Table XII
NPC-DISC-1 Disclosure Gate	`src/nesy_game/disclosure.py`; `research/npc-disclosure/reveal-hint-expectations-v1.json`, `results-signal-explainer-20261023-r5/`	60 declaration-derived `reveal_hint` cases (48 normal/4 boundary/8 malformed) across 3 arms (60/60 `general_rule` vs. `ontology_gate` parity; 51 unsafe in unchecked arm) and 2 rule-change fixtures.	`scripts/generate_disclosure_paper_results.py` Sec. IV-B1, VI, Tables XIII–XV, Fig. 4
Native Pilot & **ID Extension**	`research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json`; `codex-identity-20261026/public-results-v1/results.json`	8 assigned fixed-game rows (5/8 goals, 1 harness fault, 2 timeouts, 23 host calls, 0/16 unsafe, 7 semantic replays, 6/0/1/1 fresh-engine); plus planned 20-row identity extension stopped at Row 1 (`IDENTITY_UNVERIFIED`).	`scripts/generate_runtime_comparison_publication.py` Sec. VI, Table XVI, App. A–B
v5 Feedback & **Cohort History**	`research/codex_feedback_pilot/public-v5/`; `paired-model-comparison-20261110-v5-source.zip`; `public-history-20261111/cohort-inventory.json`	20 v5 assignments (5 fixtures $\times$ 2 selectors $\times$ Detailed/Generic; 8 gate pass, 7 goal, 10 schema fail), prompt/response SHA-256 digests, 7 frozen source files, and 19-cohort historical inventory.	`tests/test_codex_feedback_publication.py` Sec. VI, Table XVII, Figs. 6–7, App. D
Withdrawn History & **Blocked Pilot**	`research/maintenance/ollama-unverified-history-20261113.json`; `research/prospective-pilot-20261103/protocol-v1.json`, `research/prospective-pilot-20261103/results-v1/collection-status.json`, `summary.json`	Ollama attribution withdrawn: original receipts unavailable; historical JSON is not validated experimental data. Separate prospective plan: 0/180 completed (60 runtime/120 dev planned), 9 fail-closed blockers (`NOT_RUN/BLOCKED`).	`scripts/run_prospective_pilot.py` App. C–D
Crosswalk & **Manifests**	`research/academic-pipeline/*-matrix.csv`, `stage-05-citation-verification.json`; `visuals/source-manifest.json`; `research/traceability-scoring-20261023/rubric.json`	58-reference topic/status crosswalk (48 verified, 10 preprint), C1–C5 & E1–ENG1 claim-boundary matrices, visual/table SHA-256 source manifest, and 6-dimension traceability rubric (`TR01--TR06`).	`scripts/validate_visual_assets.py` Table I, Table V, Supplement